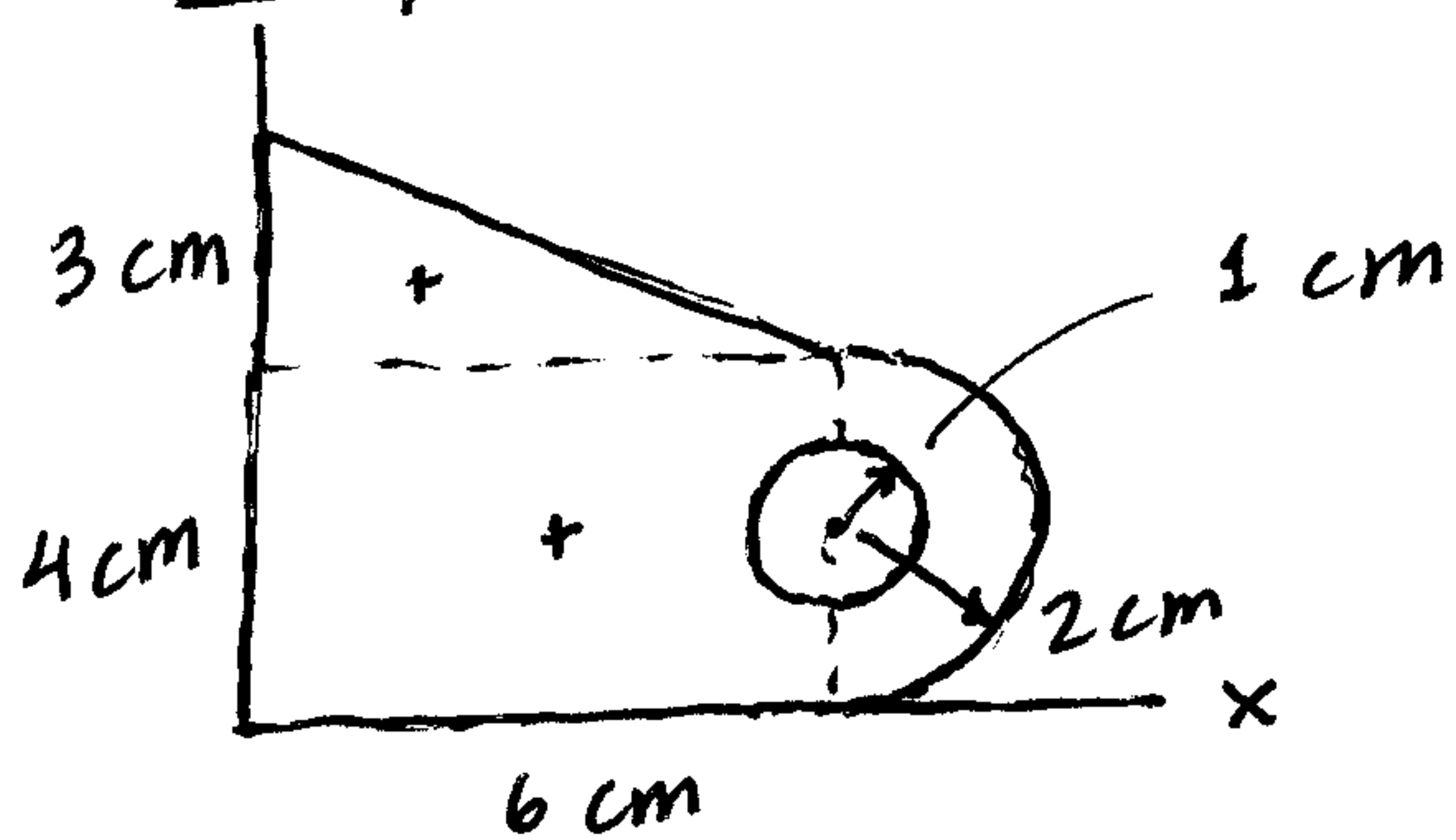


I_{xy} Example Problem 3



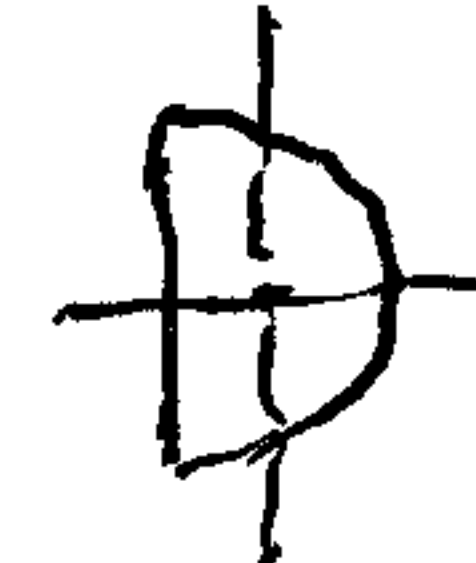
We already determined:

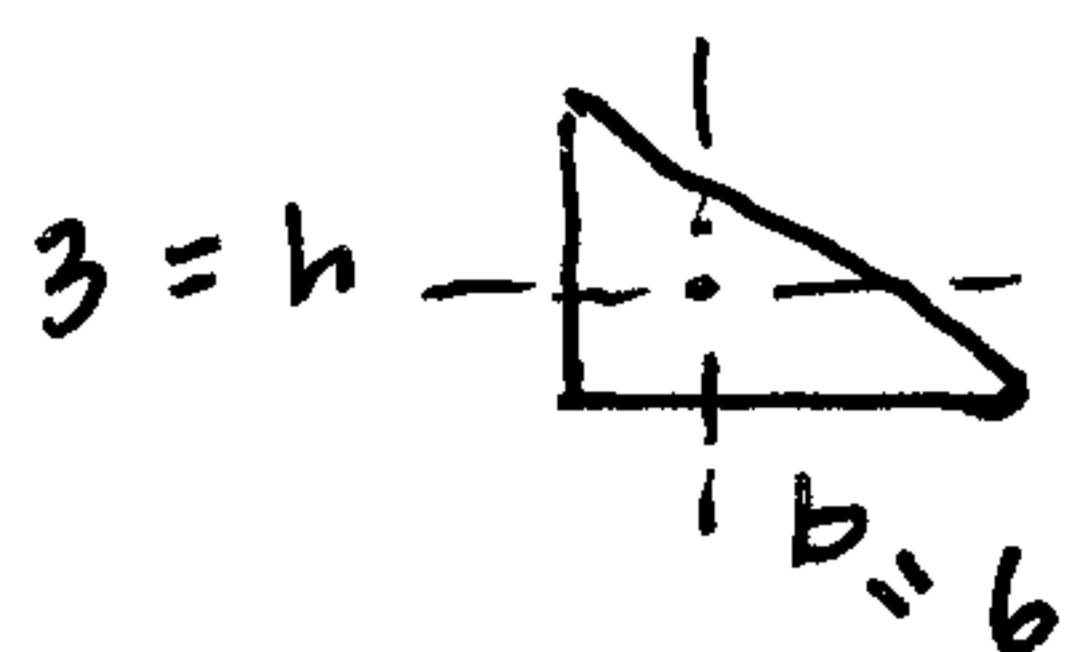
$$I_x = 398.1 \text{ cm}^4$$

$$I_y = 524.6 \text{ cm}^4$$

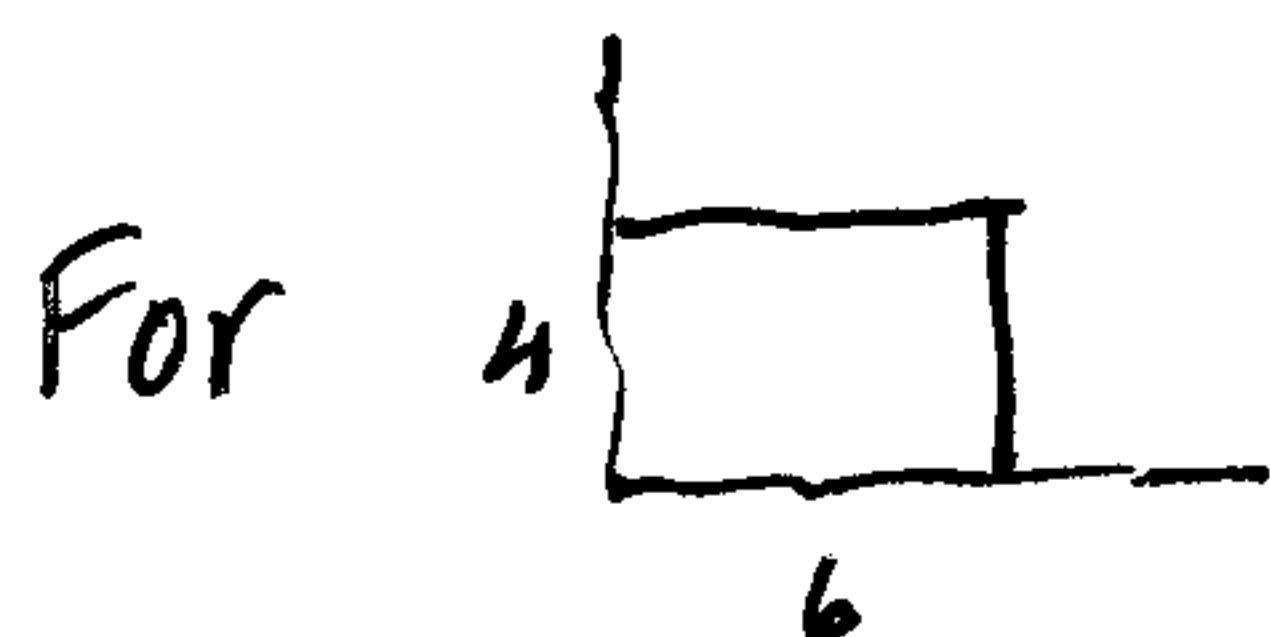
Determine I_{xy} .

$$I_{xy} = \sum (\bar{I}_{x'y'} + A d_x d_y)$$

Notes: $\bar{I}_{x'y'} = 0$ for , , 



$$\bar{I}_{x'y'} = -\frac{1}{12} b^2 h^2 = -\frac{1}{12} (6)^2 (3)^2 = -4.5 \text{ cm}^4$$



For  can use $I_{\text{origin}} = +\frac{1}{4} (6)^2 (4)^2 = 144 \text{ cm}^4$

$$I_{xy} = [+144]_{\square} + \left[-4.5 + \underbrace{\frac{1}{2}(6)(3)}_A \underbrace{(2)}_{d_x} \underbrace{(4+1)}_{d_y} \right]_{\triangle}$$

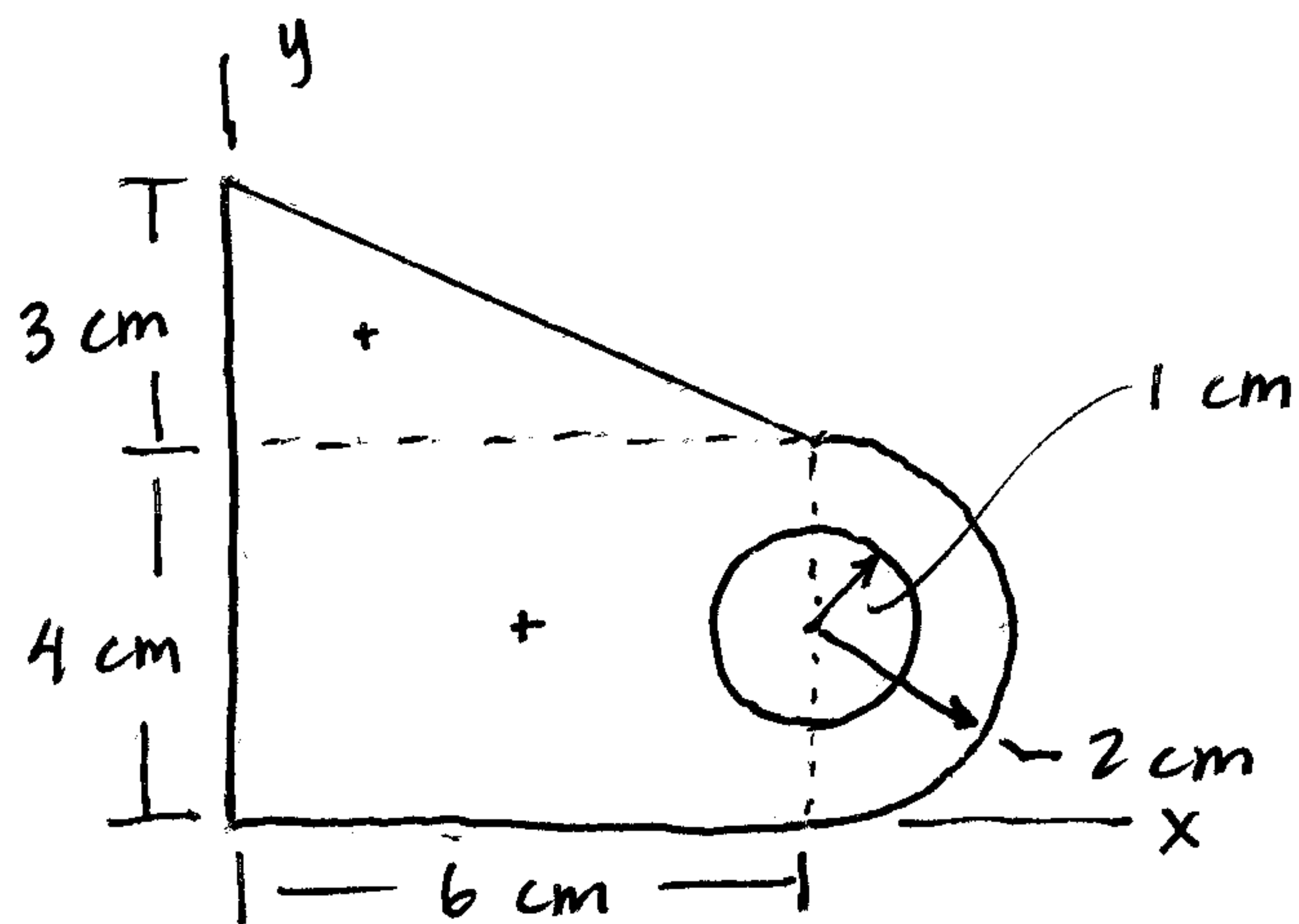
$$+ \left[0 + \frac{\pi(2)^2}{2} \left(6 + \frac{4(2)}{3\pi} \right) (2) \right]_D - \left[0 + \pi(1)^2(6)(2) \right]$$

6.8488

$$I_{xy} = 144 + 85.5 + 86.1 - 37.7$$

$$I_{xy} = 277.9 \text{ cm}^4$$

Composite Area MOI Ex Problem 3



Determine I_x, I_y

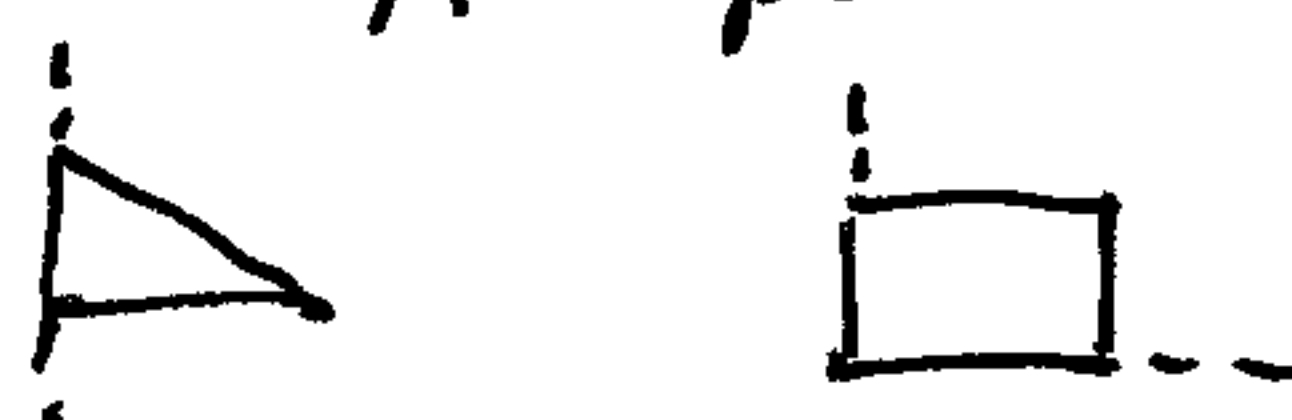
PAT: $I_{Axis} = \sum (\bar{I} + Ad^2)$

Think: 4 shapes...

$\square$, $\triangle$, D , $\bigcirc$ Hole!

* Subtract the hole!

* Easier to use I_{base} formulas if possible...



$$I_x = \left[\frac{1}{3}(6)(4)^3 \right]_{\square} + \left[\frac{1}{36}(6)(3)^3 + \frac{1}{2}(6)(3)(4+1)^2 \right]_{\triangle} + \left[\frac{\pi}{8}(2)^4 + \frac{\pi}{2}(2)^2(2)^2 \right]_D - \left[\frac{\pi}{4}(1)^4 + \pi(1)^2(2)^2 \right]_{\bigcirc}$$

↑ subtract!

$$I_x = 128 + [27 + 225] + [6.283 + 8\pi] - \left[\frac{\pi}{4} + 4\pi \right]$$

$$I_x = 398.1 \text{ cm}^4$$

$$I_y = \left[\frac{1}{3}(4)(6)^3 \right]_{\square} + \left[\frac{1}{12}(3)(6)^3 \right]_{\triangle}$$

$$+ \left[.1098(2)^4 + \frac{\pi(2)^2}{2} \left(6 + \frac{4(2)}{3\pi} \right)^2 \right]$$

$$- \left[\frac{\pi}{4}(1)^4 + \pi(1)^2(6)^2 \right]$$

$$= 288 + 54 + [1.757 + 2\pi(6.849)^2] - \left[\frac{\pi}{4} + 36\pi \right]$$

$$= 288 + 54 + 1.757 + 294.7 - \frac{\pi}{4} - 36\pi$$

$$I_y = 524.6 \text{ cm}^4$$